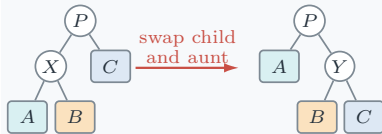


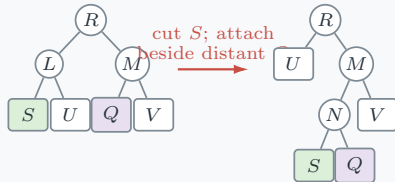
(a) **PERCH**: nearby rotation



One internal edge changes.
PERCH selects the swap
by distance and balance.

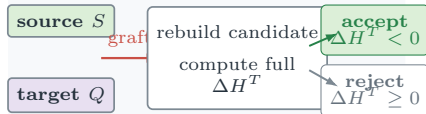
$$((A, B), C) \longrightarrow (A, (B, C))$$

(b) **GRINCH**: distant graft



A whole subtree moves across the tree.
GRINCH selects the distant target
by linkage similarity.

(c) **NEST-G**: checked graft (proposal)



Use a graft for long-range repair.
Commit it only if structural entropy decreases.